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Statistical Causal Inference

Emerging Paradigms for Advanced Data Analysis

Julian **Frattini**, Ph.D.
Chalmers University of Technology, Sweden
https://julianfrattini.github.io/

JulianFrattini/**bda4sci** Public

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Julian Frattini // about me

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2013 - 2019 **Bachelor & Master of Science**

Technical University of Munich, Munich, Germany

2020 – 2025 **Ph.D. Student**

Blekinge Institute of Technology, Karlskrona, Sweden

2025 - 2029 **Postdoctoral Researcher**

Chalmers University of Technology, Gothenburg, Sweden

Julian Frattini // about my research

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Good-enough Requirements Engineering

Quantifying the effect of quality factors of requirements artifacts on downstream software development activities

Hybrid-AI Software Engineering

Understanding the interface between traditional and data-driven software components in a hybrid software system.

Software Engineering Research Methodology

Improving the methods used in research to produce more rigorous and valid conclusions

Motivation

The Point of Science

State-of-the-art

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We would like to
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on understanding the
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issue tracking systems
measures. Overall, our
picture of the relations
issue discussions have
influenced.

Sanei, A., Cheng, J., & Adams, B. (20
generated issue discussions. In *202*
Human Aspects of Software Enginee
10.1109/CHASE52884.2021.00009

1 Introduction

Software requirements continue to evolve during software development and maintenance, and the associated risk to cost, project schedule, and quality appeals to the need for increased understanding of the phenomena. The case studies presented in this paper are part of a family of collaborative empirical initiatives, each of which makes progress towards the ultimate goal of requirements change anticipation. Loconsole [1] provides a model to predict requirements change which utilises the observed correlation between the size of a requirements use case document and volatility. While correlation is significant, there is no claim made with respect to causality. Our approach is to use case studies to increase our understanding of the causes and effects of requirements change in order to provide an empirically founded theory upon which to base causal models. To this end, it is necessary to firstly determine an

McGee, S., & Greer, D. (2012). Towards an understanding of the causes and effects of software requirements change: two case studies. *Requirements Engineering*, *17*, 133-155. DOI: 10.1007/s00766-012-0149-0
Referring to Loconsole, A., & Borstler, J. (2005, December). An industrial case study on requirements volatility measures. In *12th Asia-Pacific Software Engineering Conference (APSEC'05)* (pp. 8-pp). IEEE. DOI: 10.1109/APSEC.2005.38

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5.5 Threats to validity

External validity The generalizability of results cannot be claimed based on the exploratory case study on one data set. Further data sets are necessary to be investigated and compared to compensate for context factors like the size and domain of the company, the utilized development process, techniques employed in the requirements engineering phase, as well as applied technologies. However, the analyzed data-set represents five years of product development (between 2010 and 2015) with 41 products and 36 software releases. Therefore, the heterogeneity of authors and editors of requirements is high.

Internal validity To ensure that the correlation between the occurrence of causality and the lead-time of requirements is indeed causal and not confounded, further qualitative analysis beyond the data recorded in the respective data set must be performed. The impact of causality on comprehensibility in contrast to other factors of ambiguity has to be addressed in future studies of qualitative nature. Apart from

Frattini, J., Fischbach, J., Mendez, D., Unterkalmsteiner, M., Vogelsang, A., & Wnuk, K. (2023). Causality in requirements artifacts: prevalence, detection, and impact. *Requirements Engineering*, *28*(1), 49-74. DOI: 10.1007/s00766-022-00371-x

Correlation versus Causality

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Correlation:
when we *observe* X,
we also observe Y

Causality:
when we *do* X,
we observe Y.

If we want to make a **positive impact** on the target audience of our research (i.e., SE practitioners), we need to provide **reliable decision-making policies** – which we can only derive from **causal** relationships, not correlations.

> We cannot infer causality from data alone – we must also know the process that created that data.[1]

[1]we'll revisit this statement later

Experimental vs. Observational Studies

One ”straightforward” way to distinguish correlation from causality: **conduct a controlled experiment**, where we exert control over a treatment and isolate its effect from all confounding.

Expensive
Recruiting eligible participants is difficult

Unrealistic
Experiments perturb the natural context

Impossible
Some factors are impossible to control

Assumed
The validity of experimental results depends on several conditions

Causal Modelling

Framing Causal Assumptions

Causal Modeling

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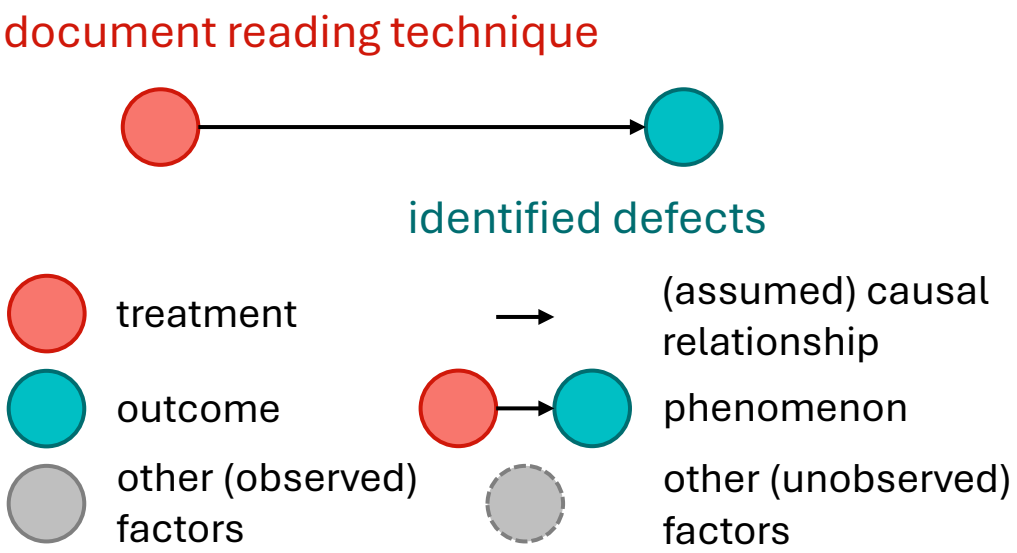
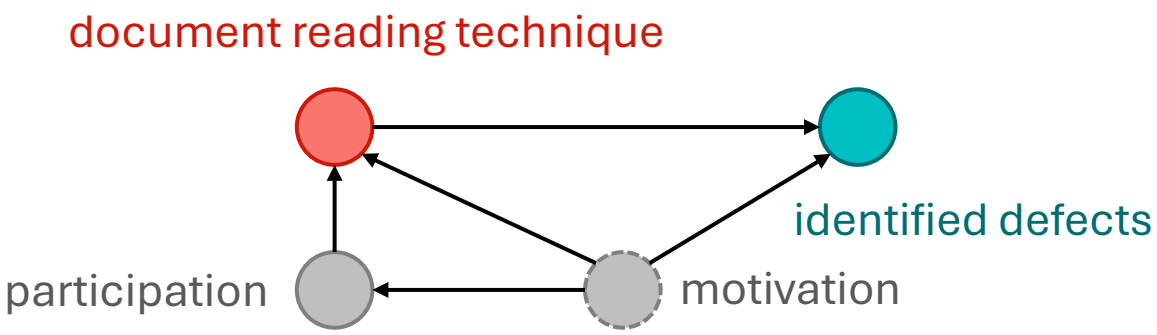
Directed, acyclic graphs (DAGs) can represent causal assumptions, where *nodes represent variables* and *edges represent assumed/potential causal relationships* between those variables.





> DAGs are subjective and represent the view of a domain expert.

Exercise: Formulating a DAG

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Derive a causal DAG from the following description:

“Researchers are interested in knowing if using ChatGPT affects the quality and length of code reviews at a company. Since they cannot interrupt the regular workflow, they cannot force anyone to use or not use ChatGPT. They do know, however, that younger developers use ChatGPT more often. On the other hand, they know that more skilled developers generally produce more concise (i.e., shorter) and better code reviews.”

Solution: Formulating a DAG

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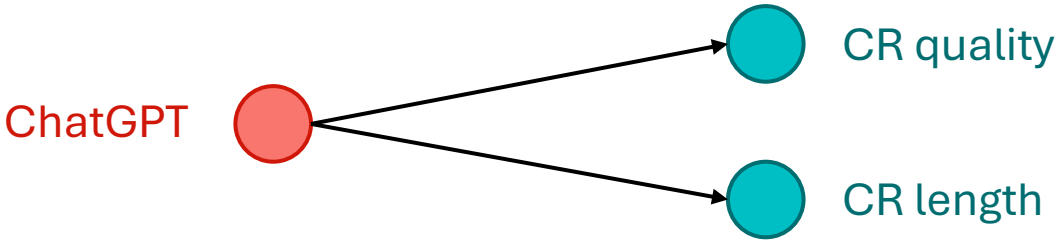
"Researchers are interested in knowing if **using ChatGPT** affects the **quality** and **length of code reviews** at a company. Since they cannot interrupt the regular workflow, they cannot force anyone to use or not use ChatGPT. They do know, however, that younger developers use ChatGPT more often. On the other hand, they know that more skilled developers generally produce more concise (i.e., shorter) and better code reviews."



Solution: Formulating a DAG

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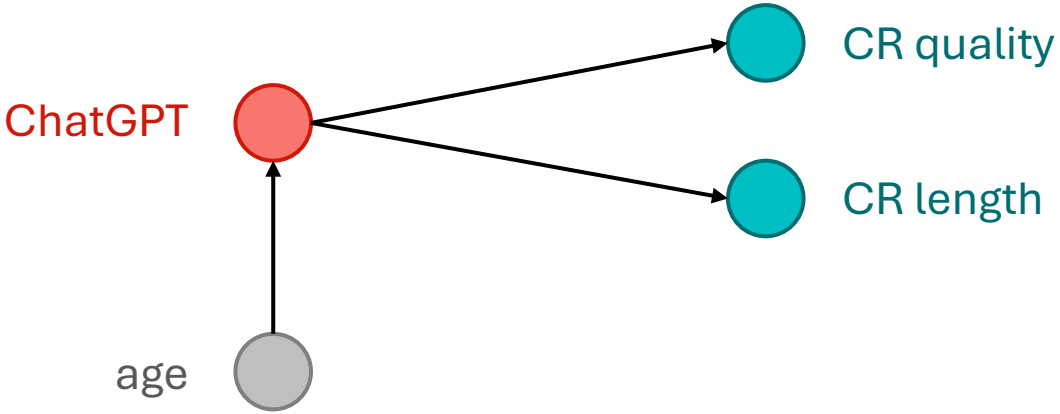

Solution: Formulating a DAG

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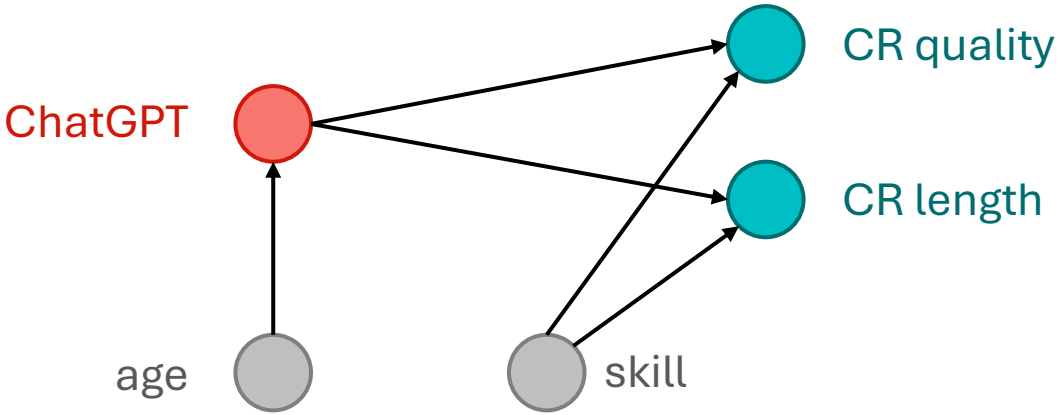

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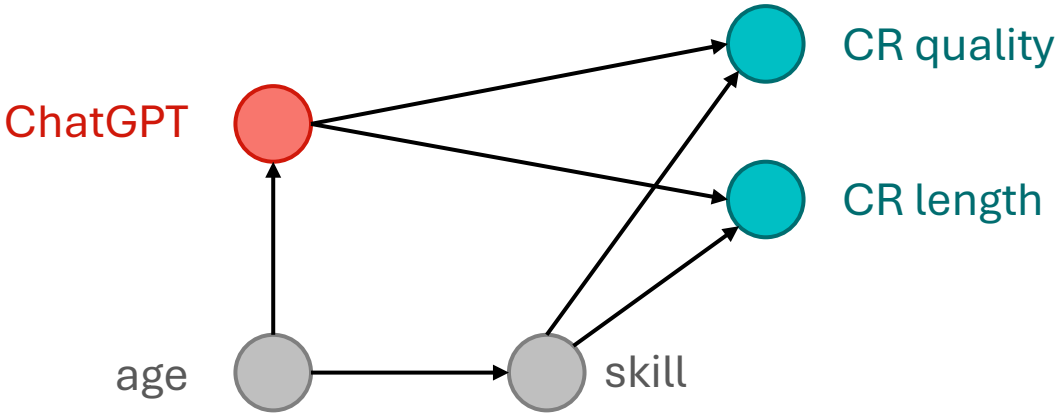

Types of Associations

Relationships Between Three (or more) Variables

Three Types of Associations

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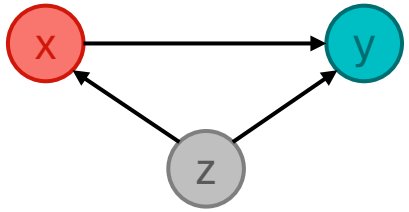
Confounder

(or: fork)

z is a common cause of both exposure x and the outcome y

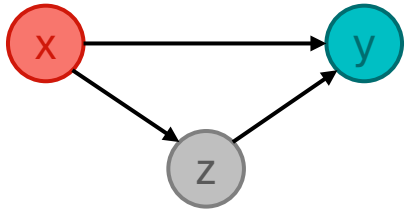


To get an unbiased estimation of the effect x → y we must control z.

Mediator

(or: pipe)

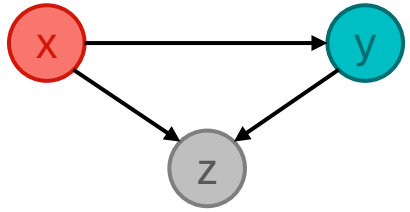
z is an effect of exposure x and a cause the outcome y



To get the direct effect x → y we must control z. To get the total effect, we must not control z.

Collider

z is a common effect of both exposure x and the outcome y



Colliding: Common Effects

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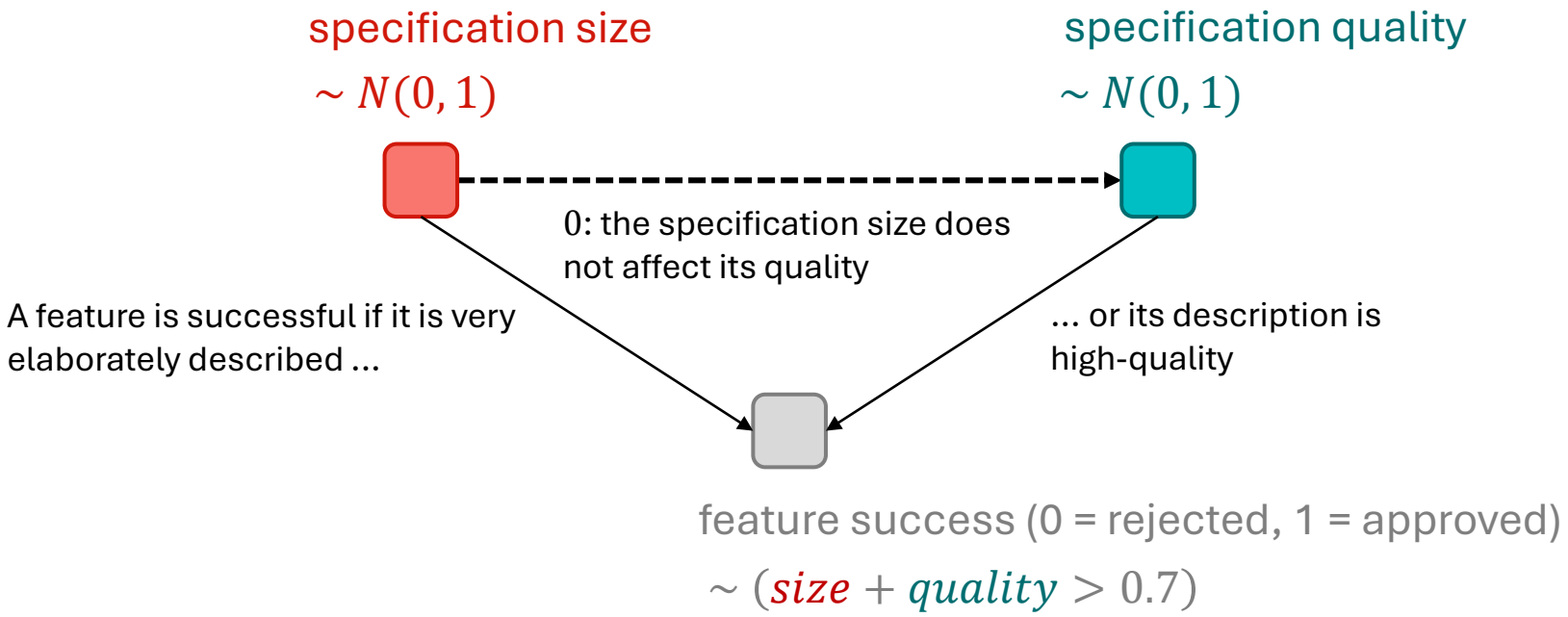
"Does the length of a requirements specification affect its quality? Both factors do affect whether the specified feature is implemented successfully."



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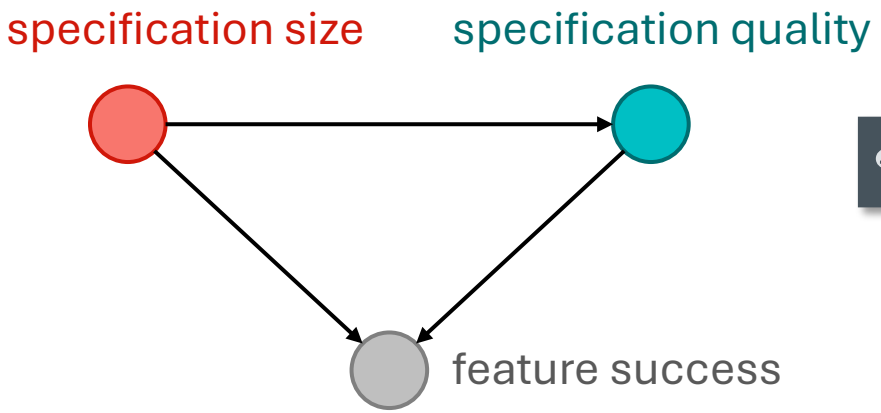
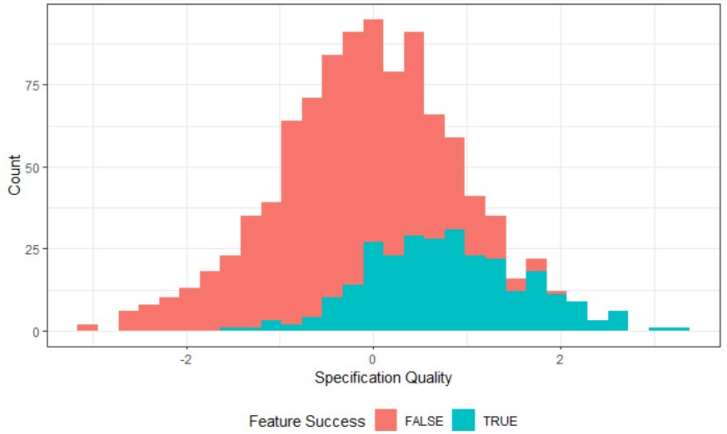


Model Derivation

m_1: $quality \sim 1 + size + success$

m_2: $quality \sim 1 + size$



Colliding: Common Effects

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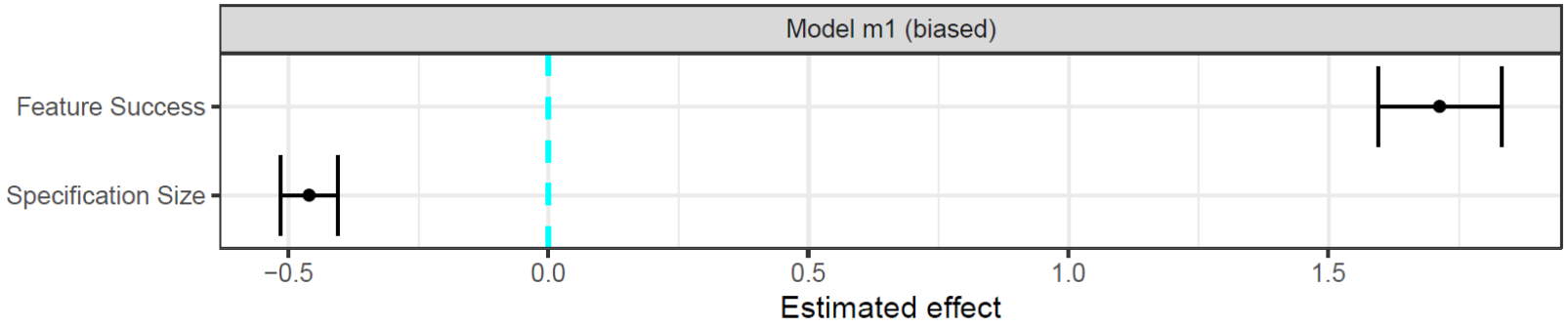

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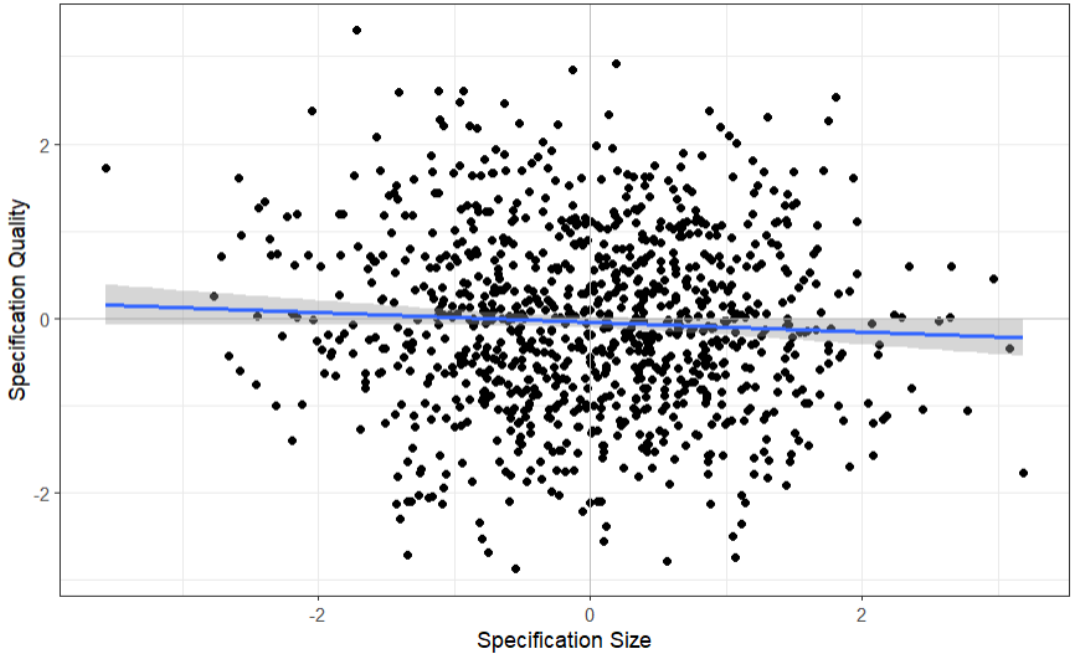

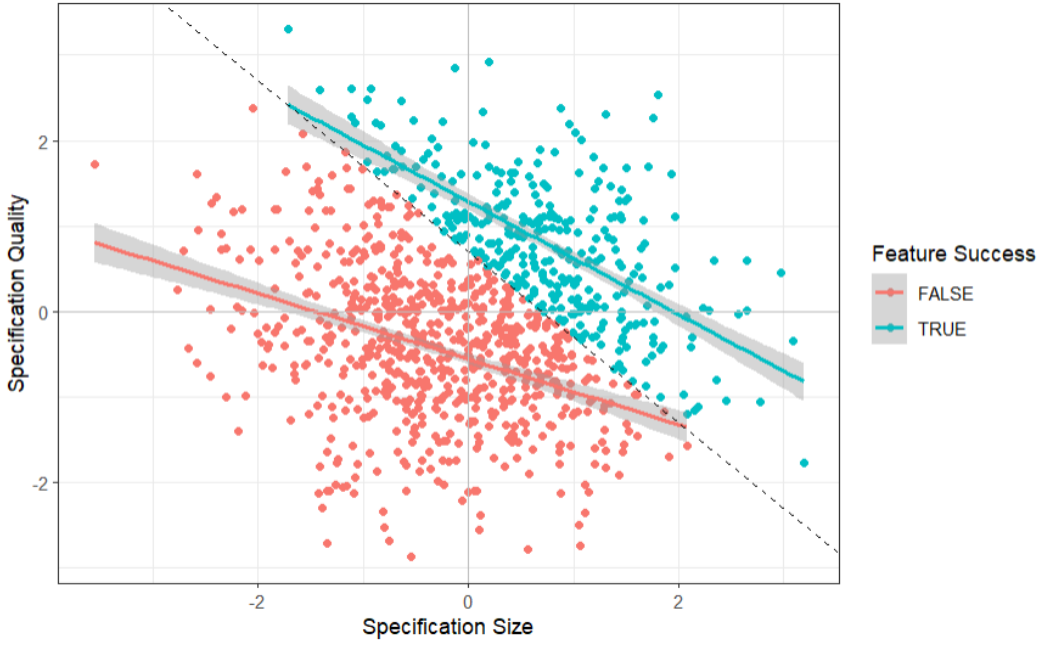

Colliding: Common Effects

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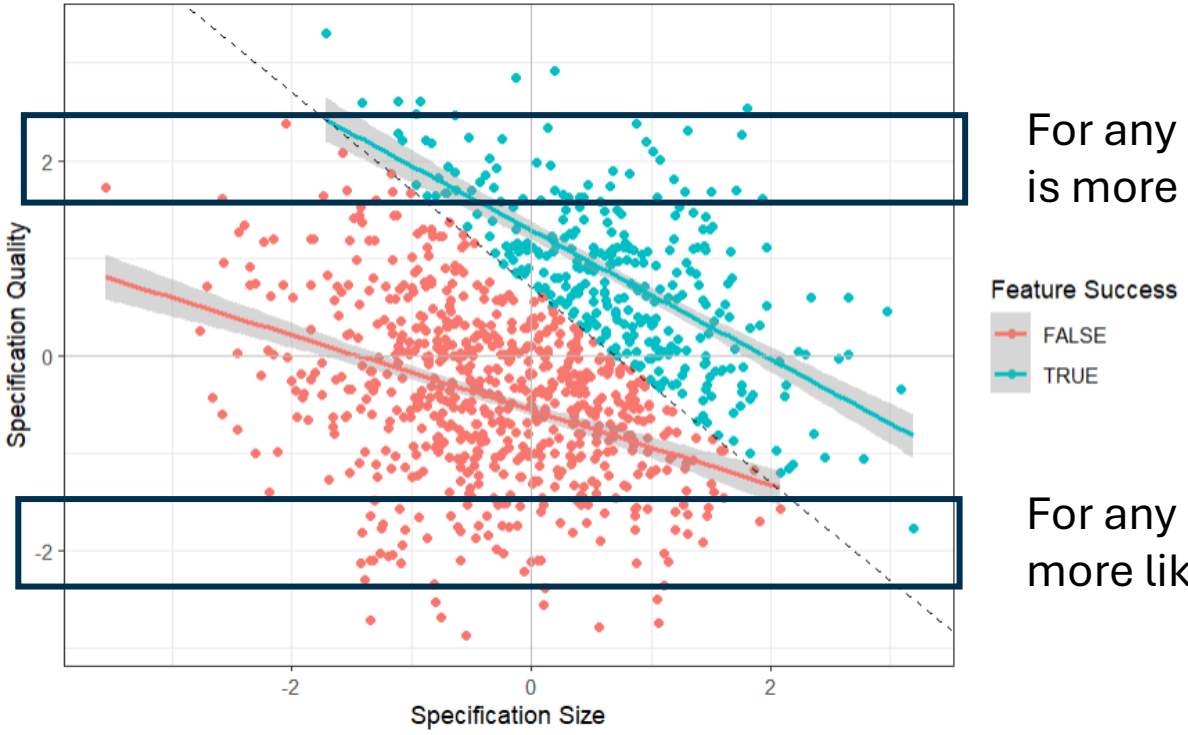

For any high-quality value, if feature success is *false*, then it is more likely that the specification size was small.

For any low-quality value, if feature success is *true*, then it is more likely that the specification size was large.

Colliding: Common Effects

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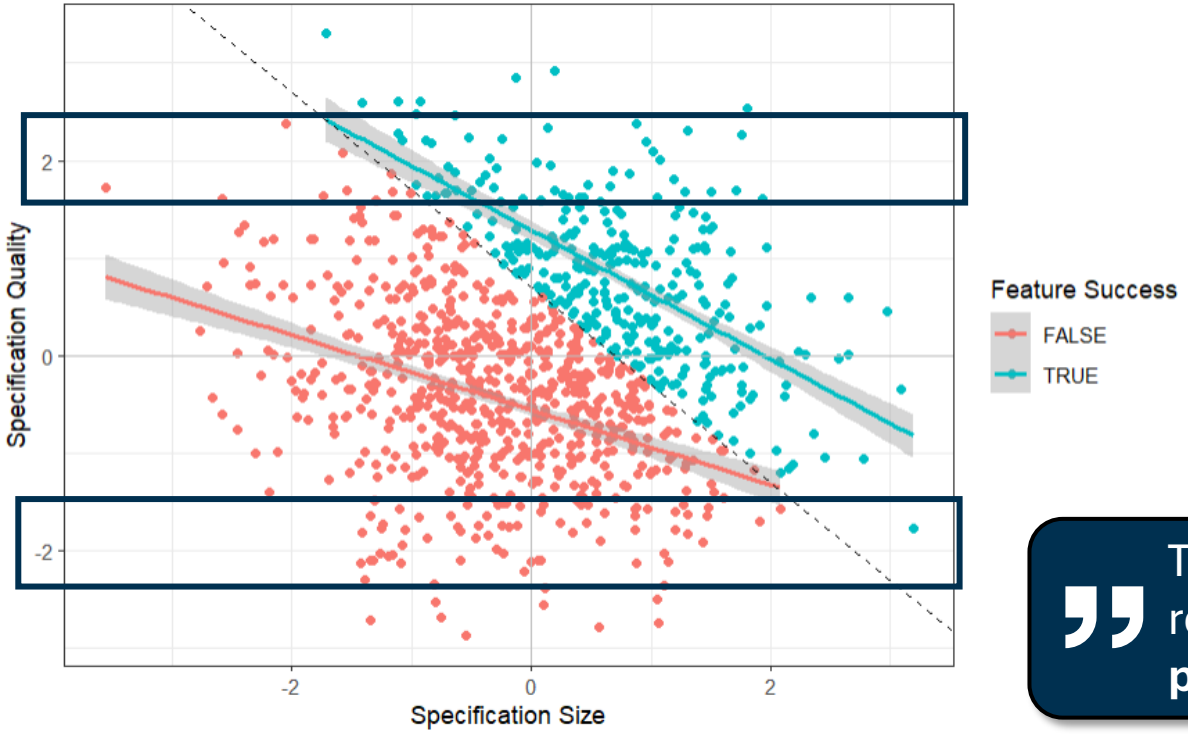

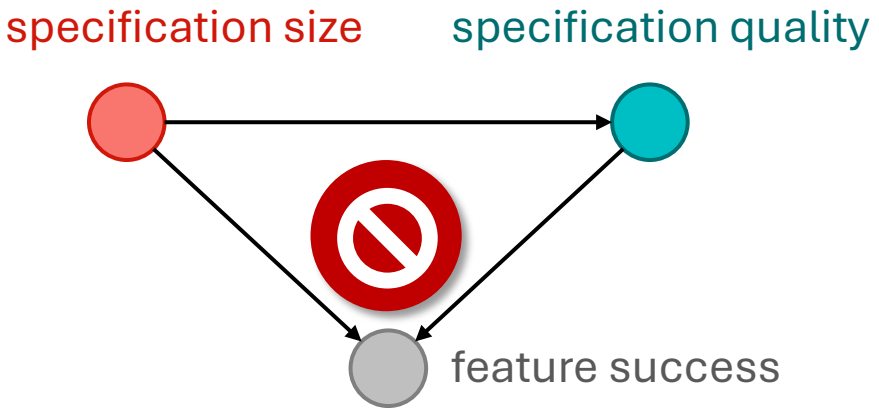

> The non-causal association suggests a negative relationship between the cause and effect (**Berkson's paradox/fallacy**).

Colliding: Common Effects

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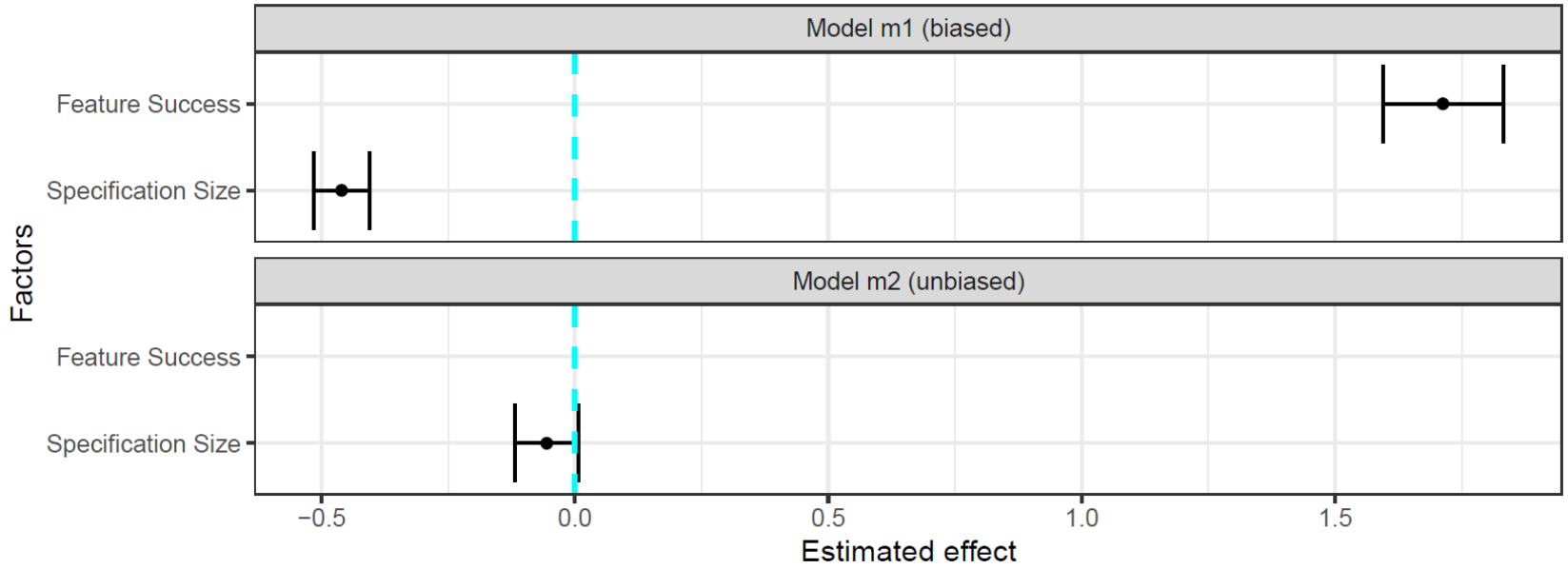

$m_1: quality \sim 1 + size + success$

$m_2: quality \sim 1 + size$

Colliding: Common Effects

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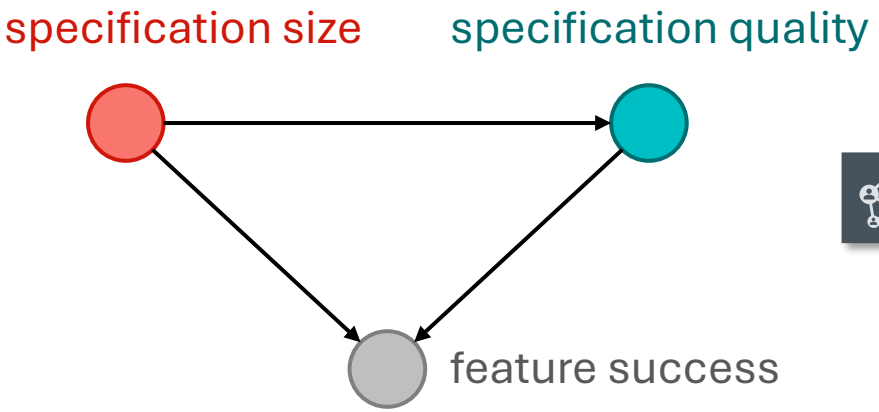


Model Derivation

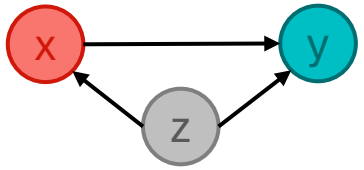
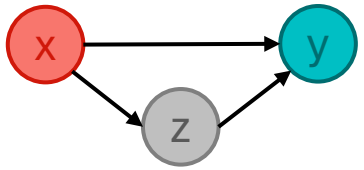
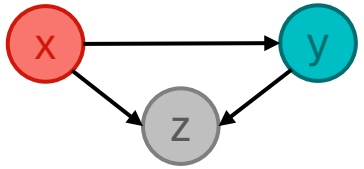
$\cancel{m_1: quality \sim 1 + size + success}$

$m_2: quality \sim 1 + size$

> If both treatment and outcome have a common effect, we must not condition on this common effect. Consequently, we cannot simply add *all* relevant variables as predictors.

Basic Types of Association

The three basic types of association behave differently when statistically controlling for the third variable:

	Not controlling z	Controlling z
 **Confounder**	✓	✗
 **Mediator**	✓	✗
 **Collider**	✗	✓

✓ Information flows from x to y

✗ Information does not flow from x to y

Identification

Causal Inference Based on Causal DAGs

Deriving a statistical Model from a causal Model: d-separation

A general framework to derive statistical models from causal models uses **d-separation**. It produces an **adjustment set**, i.e., the set of variables to control statistically in order to de-confound the causal effect between the exposure and the outcome.

https://bayes.cs.ucla.edu/BOOK-2K/d-sep.html

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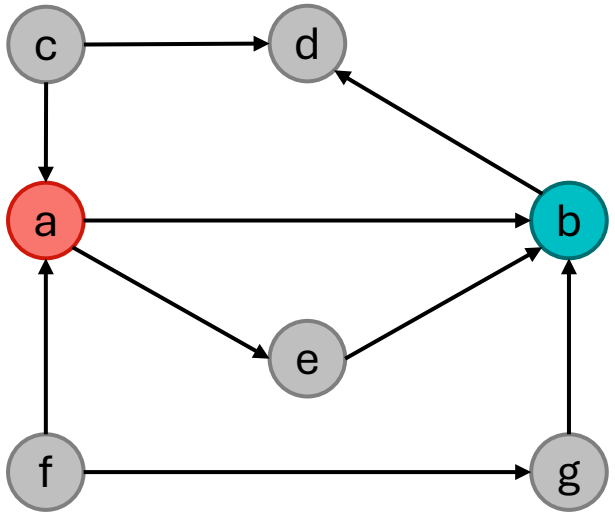

Model Derivation

$m: b \sim a$

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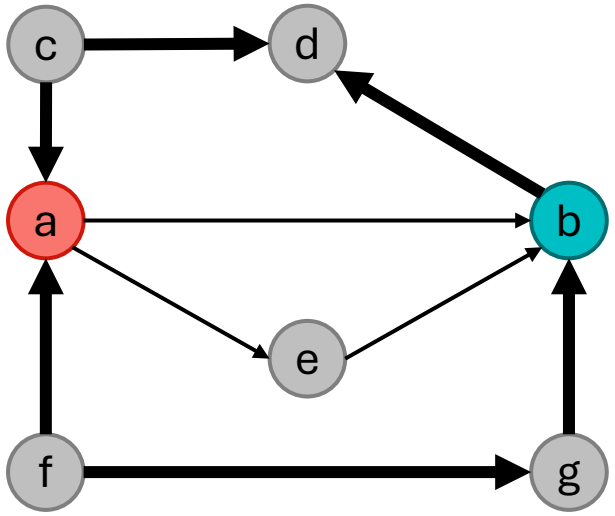

Model Derivation

$m: b \sim a$

1. Identify every **backdoor path**.

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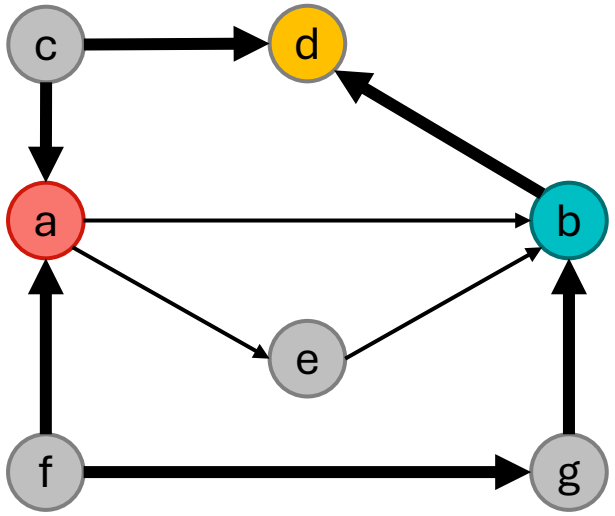

Model Derivation

$m: b \sim a$

1. Identify every backdoor path.
2. Check if there is a **collider** between the cause and effect of interest.
   1. If yes, do not condition on any of the variables on this path.

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Deriving a statistical Model from a causal Model: d-separation

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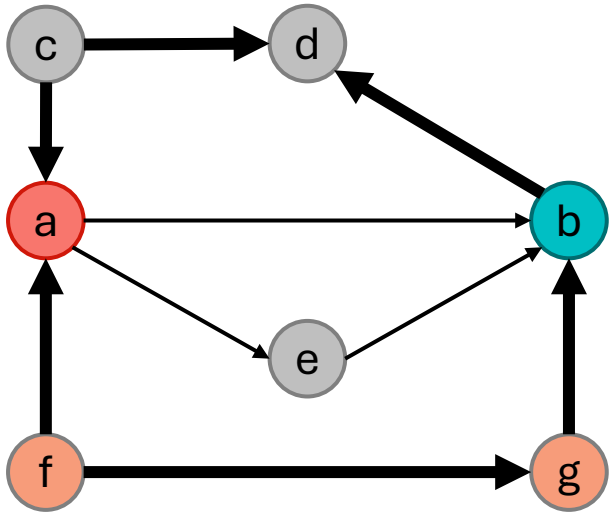

Model Derivation

$m: b \sim a + f$

$m: b \sim a + g$

1. Identify every backdoor path.
2. Check if there is a collider between the cause and effect of interest.
   1. If yes, do not condition on any of the variables on this path.
   2. If not, condition on **any of the variables** on this path to "close" the backdoor path.

https://bayes.cs.ucla.edu/BOOK-2K/d-sep.html

Deriving a statistical Model from a causal Model: d-separation

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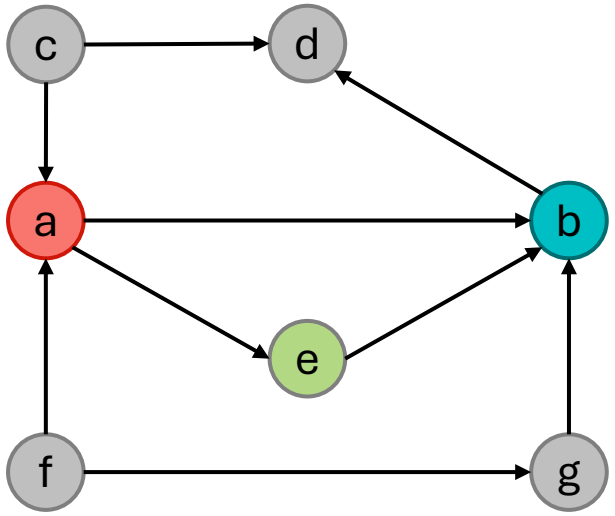


Model Derivation

$m: b \sim a + f(+e)$

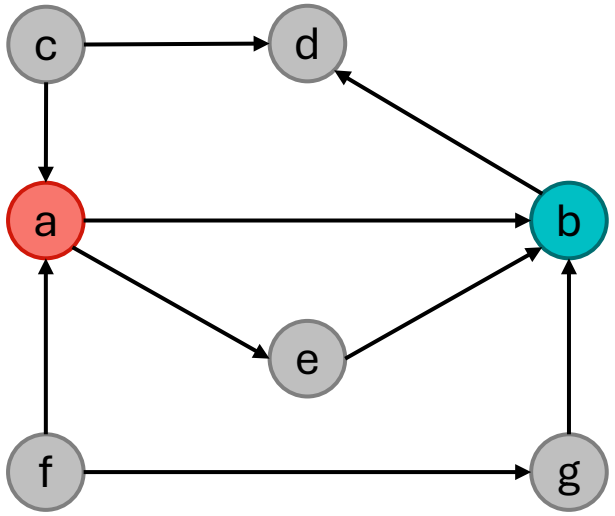
$m: b \sim a + g(+e)$

1. Identify every backdoor path.
2. Check if there is a collider between the cause and effect of interest.
   1. If yes, do not condition on any of the variables on this path.
   2. If not, condition on any of the variables on this path to "close" the backdoor path.
3. Optionally, condition on **mediators** to discern the direct effect.

https://bayes.cs.ucla.edu/BOOK-2K/d-sep.html

Deriving a statistical Model from a causal Model: d-separation

A general framework to derive statistical models from causal models uses **d-separation**. It produces an **adjustment set**, i.e., the set of variables to control statistically in order to de-confound the causal effect between the exposure and the outcome.



Model Derivation

$m: b \sim a + f(+e)$

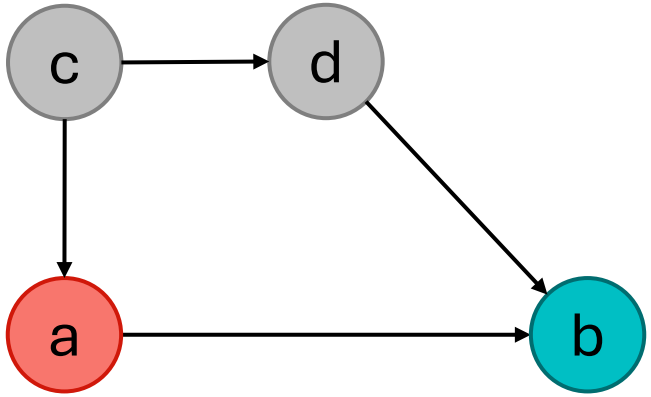
$m: b \sim a + g(+e)$

1. Identify every backdoor path.
2. Check if there is a collider between the cause and effect of interest.
   1. If yes, do not condition on any of the variables on this path.
   2. If not, condition on any of the variables on this path to "close" the backdoor path.
3. Optionally, condition on mediators to discern the direct effect.

https://bayes.cs.ucla.edu/BOOK-2K/d-sep.html

Exercises

Determine the **adjustment set** of variables to de-confound the estimation of the effect of the phenomenon a → b via d-separation.



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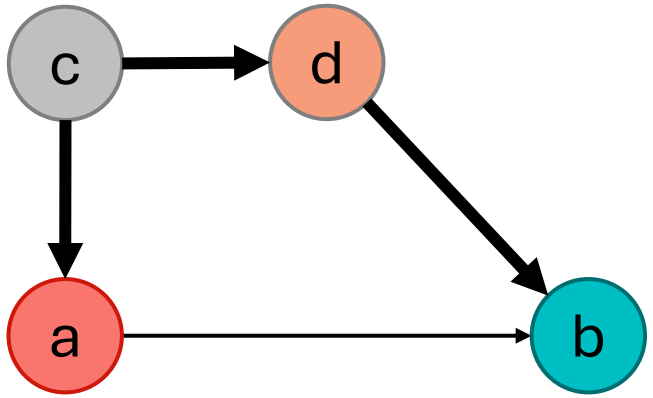
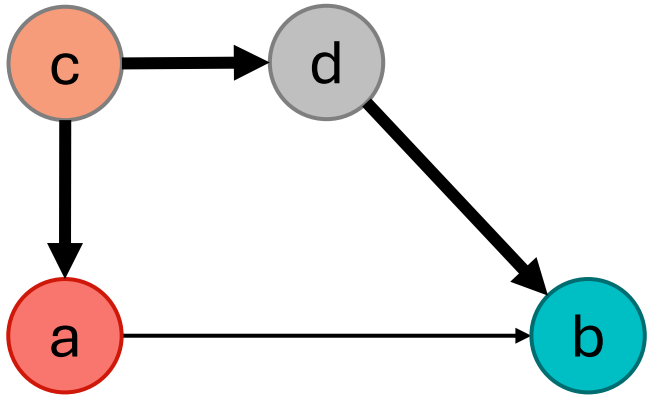


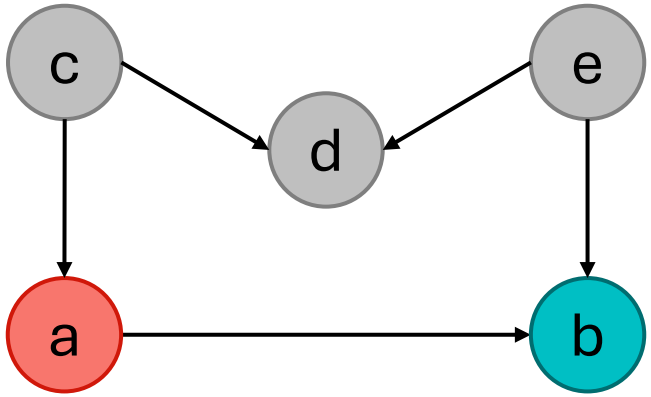


Adjustment set = {d}

Adjustment set = {c}

Exercises

Determine the **adjustment set** of variables to de-confound the estimation of the effect of the phenomenon a → b via d-separation.



Exercises

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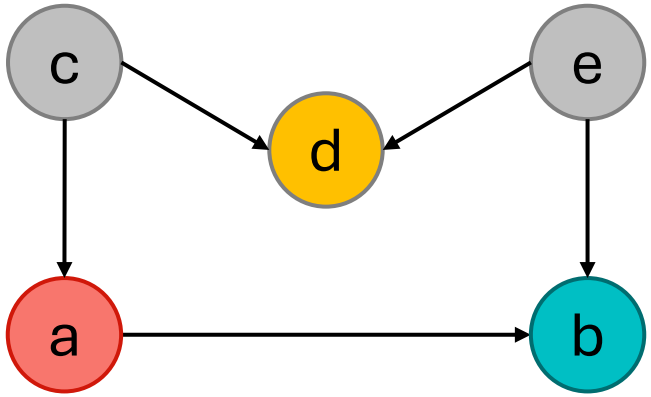

Adjustment set = {}

Exercises

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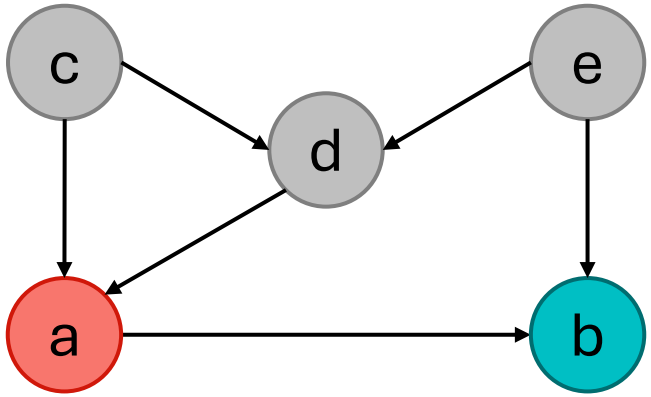

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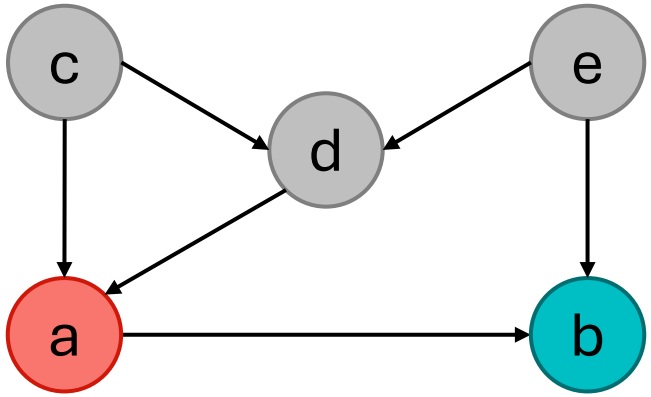
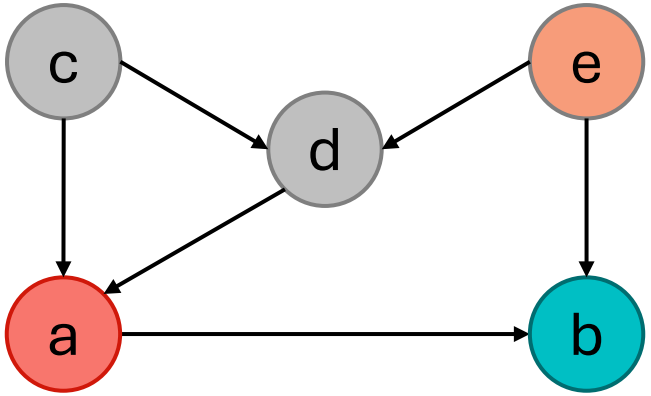


The DAG contains two backdoor paths:

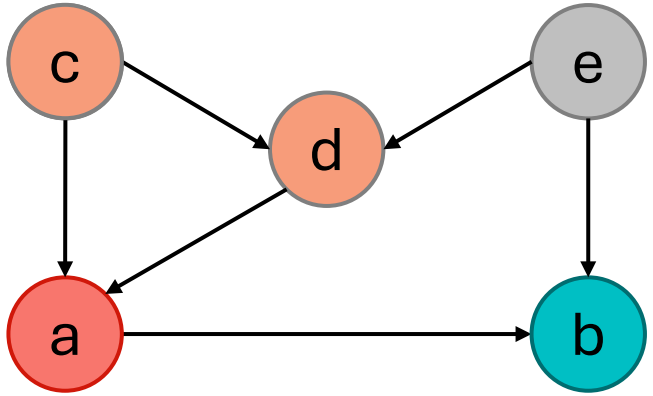
1. a ← c → d ← e → b
2. a ← d ← e → b

Exercises

Determine the **adjustment set** of variables to de-confound the estimation of the effect of the phenomenon a → b via d-separation.



Adjustment set = {e}

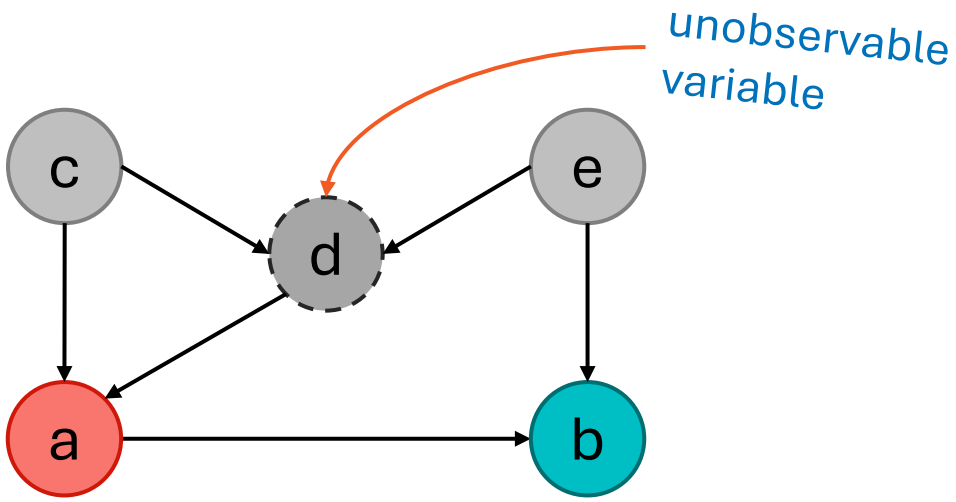


Adjustment set = {c, d}

> Conditioning on d opens the backdoor-path a ← c → d ← e → b which then has to be closed by additionally conditioning on c.

Exercises

Determine the **adjustment set** of variables to de-confound the estimation of the effect of the phenomenon a → b via d-separation.



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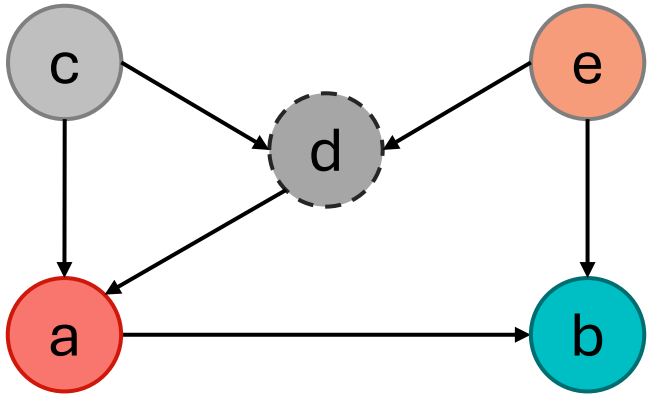

Adjustment set = {e}

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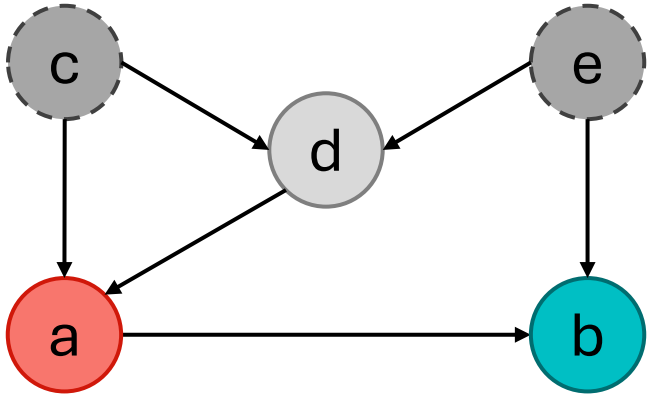

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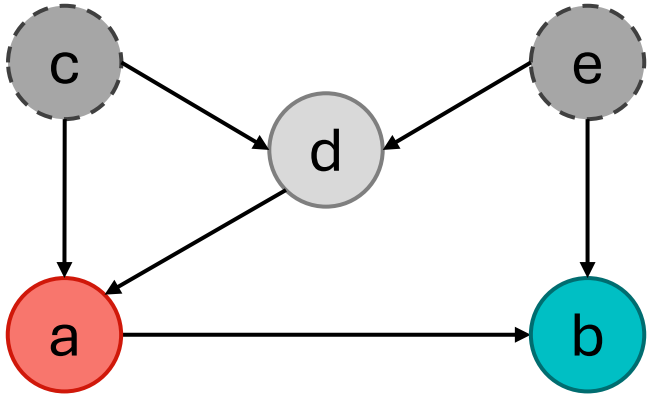

> For some DAGs, it is impossible to find an adjustment set that deconfounds the effect of interest.

Conclusion

Implications for Better Science

Internal vs. External Validity

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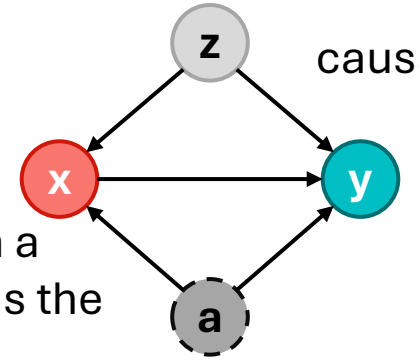
> Wohlin, C., Runeson, P., Höst, M., Ohlsson, M. C., Regnell, B., & Wesslén, A. (2012). Experimentation in software engineering (Vol. 236). Berlin: Springer.
>
> Threats to external validity are conditions that limit our ability to generalize the results of our experiment to industrial practice.

An insufficiently distributed variable with a causal effect on the phenomenon threatens the **external validity** of the conclusion.



An unobserved/unobservable variable with a causal effect on the phenomenon threatens the **internal validity** of the conclusion.

> Wohlin, C., Runeson, P., Höst, M., Ohlsson, M. C., Regnell, B., & Wesslén, A. (2012). Experimentation in software engineering (Vol. 236). Berlin: Springer.
>
> Threats to internal validity are influences that can affect the independent variable with respect to causality, without the researcher's knowledge.

Workflow for Statistical Causal Inference

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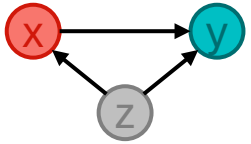
1) Modelling

Creating a DAG of assumed causal relationships



Workflow for Statistical Causal Inference

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1) Modelling

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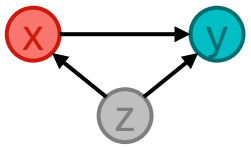


2) Identification

Deriving a statistical from the causal model

Model Derivation

$$y \sim x + z$$

3) Estimation

Training a statistical model with the observed data and evaluating the results

Siebert, J. (2023). Applications of statistical causal inference in software engineering. *Information and Software Technology*, *159*, 107198.

Scientific Workflow: Traditional

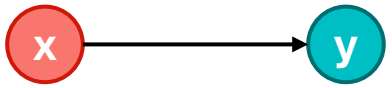
Data Collection

Modelling

Identification

Estimation

Assumed DAG



Collected Data

x	y

Too often, we derive hypotheses from the data available to us.

Scientific Workflow: Traditional

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Data Collection

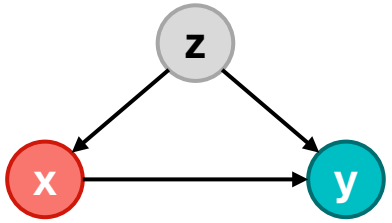
Modelling

Identification

Estimation

Assumed DAG

hard-to-measure confounder



Collected Data

x	y

> Not collecting data about a confounder will threaten the internal validity of your result

Scientific Workflow: Better

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Modelling

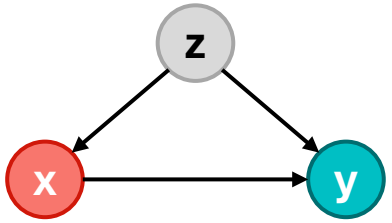
Identification

Data Collection

Estimation

Assumed DAG

hard-to-measure confounder

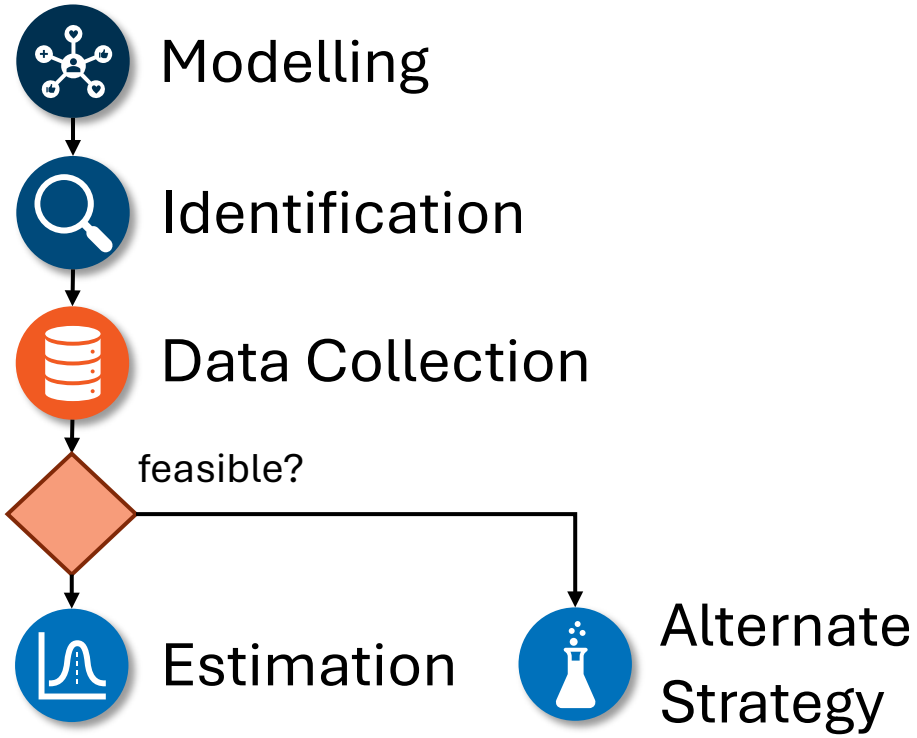


Collected Data

x	y	z

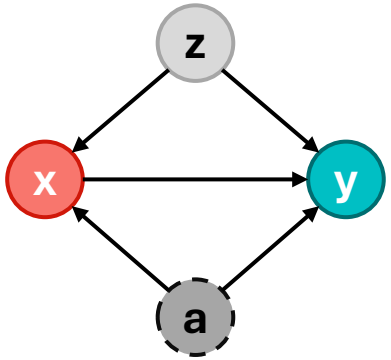
> Determine the variables about which you need to collect data in order to estimate an unbiased effect of interest *before* collecting that data.

Scientific Workflow: Best



Assumed DAG

hard-to-measure confounder



unobservable confounder

Collected Data

x	y	z	a
			X
			X
			X
			X

Unobservable confounders require (a) acknowledgement in the **threats to (internal) validity** and (b) **different research strategies**.

Coherent Science

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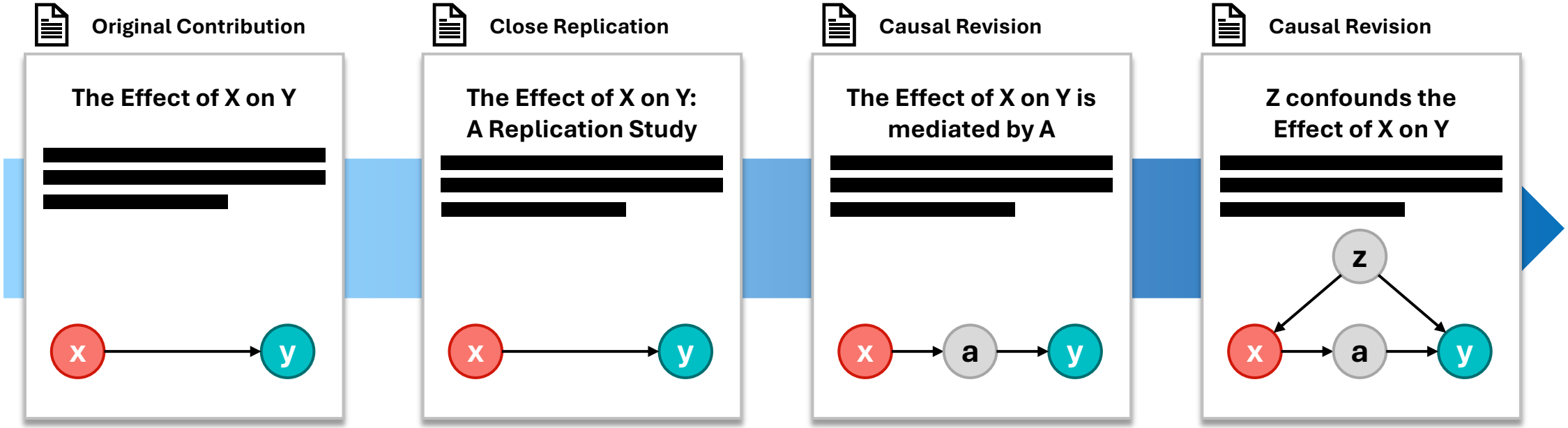
Following this scientific workflow enables us as a community to effectively contribute to a coherent scientific endeavor.



Recommendations

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Data Collection

Is a part of data analysis, not its precursor

Draw a DAG

representing your causal assumptions

Use d-separation

to figure out (1) if you can conduct your study and (2) what data you need to collect

Reading list

Pearl, J., & Mackenzie, D. (2018). *The book of why: the new science of cause and effect*. Basic books.

McElreath, R. (2018). *Statistical rethinking: A Bayesian course with examples in R and Stan*. Chapman and Hall/CRC.

Cinelli, C., **Forney**, A., & **Pearl**, J. (2024). A crash course in good and bad controls. *Sociological Methods & Research*, *53*(3), 1071-1104.

Siebert, J. (2023). Applications of statistical causal inference in software engineering. *Information and Software Technology*, *159*, 107198.

Furia, C. A., **Feldt**, R., & **Torkar**, R. (2019). Bayesian data analysis in empirical software engineering research. *IEEE Transactions on Software Engineering*, *47*(9), 1786-1810.

Furia, C. A., **Torkar**, R., & **Feldt**, R. (2022). Applying Bayesian analysis guidelines to empirical software engineering data: The case of programming languages and code quality. *ACM Transactions on Software Engineering and Methodology (TOSEM)*, *31*(3), 1-38.

Material

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All material is public and open-source on GitHub. Feel free to clone, fork, and share the study material.

Comments, requests, and additions are welcome.

JulianFrattini / bda4sci

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JulianFrattini Reorganized repository structure 9611265 · 5 months ago 23 Commits

version1	Reorganized repository structure	5 months ago
.gitignore	Reorganized repository structure	5 months ago
LICENSE	Initial commit	11 months ago
README.md	Reorganized repository structure	5 months ago
sci-intro.Rproj	Reorganized repository structure	5 months ago

README | Apache-2.0 license

Introduction to Statistical Causal Inference

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This repository contains the material for the introduction seminar to Statistical Causal Inference (SCI). The purpose of the seminar is to introduce software engineering researchers with a background in analysis of quantitative data to a causal framework for inferential statistics proposed by Judea Pearl[1] and Richard McElreath[2]

About

Material for the Introduction Seminar to Bayesian Data Analysis for Statistical Causal Inference

bayesian-data-analysis

statistical-causal-inference

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